# Donghyun Kwon

Associate Professor

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## Education

**Seoul National University** — *Seoul, Korea*
Ph.D in Department of Electrical and Computer Engineering — *Mar 2012 - Feb 2019*
- Thesis: Hardware-assisted Isolation for System Framework to Ensure Kernel Integrity
- Advisor: Yunheung Paek
- Distinguished Ph.D. Dissertation Award

**Seoul National University** — *Seoul, Korea*
B.S in Department of Electrical and Computer Engineering — *Mar 2008 - Feb 2012*

## Professional Experience

**Associate Professor** — *Pusan, South Korea*
School of Computer Science and Engineering, Pusan National University — *Mar 2023 - Present*

**Assistant Professor** — *Pusan, South Korea*
School of Computer Science and Engineering, Pusan National University — *Mar 2020 - Feb 2023*

**Senior Researcher** — *Daejeon, South Korea*
Electronics and Telecommunications Research Institute — *Mar 2019 - Feb 2020*

## Publications

** corresponding author*

1. **Mind the Gap: In-Process Isolation for Safe Rust in WebAssembly**
   Suhyeon Song, Chaewon Shin, and **Donghyun Kwon***
   European Symposium on Research in Computer Security (ESORICS), 2026
2. **XCFI: Comprehensive Control-Flow Integrity for Arm TrustZone-M**
   Yunju Gu, Jaeyeol Park, and **Donghyun Kwon***
   USENIX Security, 2026
3. **TzPy: Python Runtime for Edge Code Deployment on TrustZone**
   Suhyeon Song, Jaeyeol Park, Chaewon Shin, and **Donghyun Kwon***
   ACM Transactions on Embedded Computing Systems, 2026
4. **SERA: Secure Micro XRCE-DDS Establishment with Remote Attestation for micro-ROS**
   Jeonghwan Kang, Seungmin Kang, Euijun Kim, Jaeseon Park, Kyounghwan Kim, and **Donghyun Kwon***
   IEEE Access, 2026
5. **WARD: Efficient Memory Protection for WebAssembly on Tiny Embedded Systems**
   Chaewon Shin, Gowon Han, Suhyeon Song, Jinsun Park, Yoon-ho Choi, and **Donghyun Kwon***
   IEEE Access, 2026
6. **Beyond Address Spaces: In-process Memory Isolation for RISC-V**
   Seonghwan Park, Hayoung Kang, and **Donghyun Kwon***
   Computers & Security, 2026

7. **SMORE: Practical Redzone-Based Stack Memory Error Detection Mechanism for Embedded Systems**
Jaeyeol Park, Yunju Gu, and **Donghyun Kwon***
Annual Computer Security Applications Conference (ACSAC), 2025
8. **ELK: Effective Lock-and-Key Technique for Temporal Memory Safety on Embedded Devices in ARMv8-M**
Jeonghwan Kang, Kyounghwan Kim, and **Donghyun Kwon***
Annual Computer Security Applications Conference (ACSAC), 2025
9. **Watch Your Callback: Offline Anomaly Detection using Machine Learning in ROS 2**
Jeonghwan Kang, Kyounghwan Kim, and **Donghyun Kwon***
IEEE Access, 2025
10. **WasDom: An Efficient Write Protection for Wasm JITed Code With ARM Domain**
Suhyeon Song, Chaewon Shin, and **Donghyun Kwon***
IEEE Access, 2025
11. **ROSec: Intra-Process Isolation for ROS Composition with Memory Protection Keys**
Jiwon Seo, Kayondo Martin, Jeonghwan Kang, **Donghyun Kwon***, and Yunheung Paek*
IEEE Transactions on Automation Science and Engineering (T-ASE), 2025
12. **Non-volatile flip-flop with soft error tolerant capability using DICE and C-element**
Shogo Takahashi, **Donghyun Kwon**, and Kazuteru Namba*
IEICE Nonlinear Theory and Its Applications, 2024
13. **Look Before You Access: Efficient Heap Memory Safety for Embedded Systems on ARMv8-M**
Jeonghwan Kang, Jaeyeol Park, Jiwon Seo, and **Donghyun Kwon***
Design Automation Conference (DAC), 2024
14. **MECAT: Memory-Safe Smart Contracts in ARM TrustZone**
Seonghwan Park, Hayoung Kang, Sanghun Han, Jonghee M. Youn*, and **Donghyun Kwon***
IEEE Access, 2024
15. **Enhancing ad Lock-and-key Scheme with MTE to Mitigate Use-After-Frees**
Inyoung Bang, Martin Kayondo, Junseung Yu, **Donghyun Kwon**, Yeongpil Cho*, and Yunheung Paek*
IEEE Access, 2023
16. **metaSafer: A Technique to Detect Heap Metadata Corruption in WebAssembly**
Suhyeon Song, Seonghwan Park, and **Donghyun Kwon***
IEEE Access, 2023
17. **ZOMETAG: Zone-based Memory Tagging for Fast, Deterministic Detection of Spatial Memory Violations on ARM**
Jiwon Seo, Junseung You, **Donghyun Kwon**, Yeongpil Cho*, and Yunheung Paek*
IEEE Transactions on Information Forensics and Security (TIFS), 2023
18. **SFITAG: Efficient Software Fault Isolation with Memory Tagging for ARM Kernel Extensions**
Jiwon Seo, Junseung You, Yungi Cho, Yeongpil Cho, **Donghyun Kwon***, Yunheung Paek*
ACM Asia Computers and Communications Security (ASIACCS), 2023
19. **DEMIX: Domain-Enforced Memory Isolation for Embedded System**
Haeyoung Kim, Harashta Tatimma Larasati, Jonguk Park, Howon Kim, and **Donghyun Kwon***
Sensors, 2023
20. **Exploring Effective Uses of the Tagged Memory for Reducing Bounds Checking Overheads**
Jiwon Seo, Inyoung Bang, Yungi Cho, Jangseop Shin, Dongil Hwang, **Donghyun Kwon**, Yeongpill Cho, Yunheung Paek
The Journal of Supercomputing, 2022
21. **Efficient CFI Enforcement for Embedded Systems Using ARM TrustZone-M**
Gisu Yeo, Yeryeong Kim, Suhyeon Song, **Donghyun Kwon***
IEEE Access, 2022
22. **Bratter: An instruction set extension for forward control-flow integrity in RISC-V**
Seonghwan Park, Dongwook Kang, Jeonghwan Kang, **Donghyun Kwon***
Sensors, 2022
23. **CoMeT: Configurable Tagged Memory Extension**
Jinjae Lee, Derry Pratama, Minjae Kim, Howon Kim, **Donghyun Kwon***
Sensors, 2021

24. **A Hardware Platform for Ensuring OS Kernel Integrity on RISC-V**
**Donghyun Kwon**, Dongil Hwang, Yunheung Paek
Electronics, 2021

25. **RIMI: Instruction-level Memory Isolation form Embedded Systems on RISC-V**
Haeyoung Kim, Jinjae Lee, Derry Pratama, Asep Muhamad Awaludin, Howon Kim, **Donghyun Kwon***
International Conference On Computer Aided Design (ICCAD), 2020

26. **PrOS: Light-weight Privatized Secure OSes in ARM TrustZone**
**Donghyun Kwon**, Jiwon Seo, Yeongpil Cho, Byoungyoung Lee, Yunheung Paek
IEEE Transactions on Mobile Computing (TMC), 2020

27. **RiskiM: Toward Complete Kernel Protection with Hardware Support**
Dongil Hwang, Myunghoon Yang, Seongil Jeon, Younghan Lee, **Donghyun Kwon***, Yunheung Paek
Design, Automation and Test in Europe (DATE), 2019

28. **Safe and Efficient Implementation of a Security System on ARM using Intra-Level Privilege Separation**
**Donghyun Kwon**, Hayoon Yi, Yeongpil Cho, Yunheung Paek
ACM Transactions on Privacy and Security (TOPS), 2019

29. **CRCount: Pointer Invalidation with Reference Counting on Mitigate Use-After-Free in Legacy C/C++**
Jangseop Shin, **Donghyun Kwon**, Jiwon Seo, Yeongpil Cho, Yunheung Paek
Network and Distributed System Security Symposium (NDSS), 2019

30. **uXOM: Efficient eXecute-Only Memory on ARM Cortex-M**
**Donghyun Kwon**, Jangseop Shin, Giyeol Kim, Byoungyoung Lee, Yeongpil Cho, Yunheung Paek
USENIX Security, 2019

31. **Hypernel: A Hardware-Assisted Framework for Kernel Protection without Nested Paging**
**Donghyun Kwon****, Kuenwhee Oh**, Junmo Park, Seungyong Yang, Yeongpil Cho, Brent Byunghoon Kang, Yunheung Paek
Design Automation Conference (DAC), 2018
(**) joint first authors

32. **VM-CFI: Control-Flow Integrity for Virtual Machine Kernel Using Intel PT**
**Donghyun Kwon**, Jiwon Seo, Sehyun Baek, Giyeol Kim, Sunwoo Ahn, Yunheung Paek
International Conference on Computational Science and Its Applications (ICCSA), 2018

33. **Instruction-Level Data Isolation for the Kernel on ARM**
Yeongpil Cho, **Donghyun Kwon**, Yunheung Paek
Design Automation Conference (DAC), 2017

34. **Optimization Techniques to Enable Execution Offoading for 3D Video Games**
**Donghyun Kwon**, Seungjun Yang, Yunheung Paek, Kwangman Ko
Multimedia Tools and Applications (MTAP), 2017

35. **Dynamic Virtual Address Range Adjustment for Intra-Level Privilege Separation on ARM**
Yeongpil Cho, **Donghyun Kwon**, Hayoon Yi, Yunheung Paek
Network and Distributed System Security Symposium (NDSS), 2017

36. **Precise Execution Offloading for Applications with Dynamic Behavior in Mobile Cloud Computing**
Yongin Kwon, Hayoon Yi, **Donghyun Kwon**, Seungjun Yang, Yeongpil Cho, Yunheung Paek
Pervasive and Mobile Computing (PMC), 2016

37. **Hardware-Assisted On-Demand Hypervisor Activation for Efficient Security Critical Code Execution on Mobile Devices**
Yeongpil Cho, Jun-Bum Shin, **Donghyun Kwon**, MyungJoo Ham, Yuna Kim, Yunheung Paek
USENIX Annual Technical Conference (ATC), 2016

38. **Mantis: Efficient Predictions of Execution Time, Energy Usage, Memory Usage and Network Usage on Smart Mobile Devices**
Yongin Kwon, Sangmin Lee, Hayoon Yi, **Donghyun Kwon**, Seungjun Yang, Byung-gon Chun, Ling Huang, Petros Maniatis, Mayur Naik, Yunheung Paek
IEEE Transactions on Mobile Computing (TMC), 2015

39. **Techniques to Minimize State Transfer Costs for Dynamic Execution Offloading in Mobile Cloud Computing**
Seungjun Yang, **Donghyun Kwon**, Hayoon Yi, Yeongpil Cho, Yongin Kwon, Yunheung Paek
IEEE Transactions on Mobile Computing (TMC), 2014

40. **Fast Dynamic Execution Offloading for Efficient Mobile Cloud Computing**
Seungjun Yang, Yongin Kwon, Yeongpil Cho, Hayoon Yi, **Donghyun Kwon**, Jonghee Youn, Yunheung Paek
IEEE International Conference on Pervasive Computing and Communications (PerCom), 2013

41. **Mantis: Automatic Performance Prediction for Smartphone Applications**
Yongin Kwon, Sangmin Lee, Hayoon Yi, **Donghyun Kwon**, Seungjun Yang, Byung-Gon Chun, Ling Huang, Petros Maniatis, Mayur Naik, Yunheung Paek
USENIX Annual Technical Conference (ATC), 2013

## Academic Service

- (Reviewer) ACM Transactions on Privacy and Security (TOPS)
- (Reviewer) ACM Transactions on Architecture and Code Optimization (TACO)
- (Reviewer) IEEE Transactions on Information Forensics & Security (TIFS)
- (Reviewer) IEEE Transactions on Computers (TC)
- (Reviewer) IEEE Access
- (Reviewer) Journal of Supercomputing
- (Associate Editor) Journal of Information Processing Systems (JIPS)

## Teaching

Grad. **System Vulnerabilities and Countermeasures**, Fall 2021 - Fall 2025
Grad. **Introduction to System and Software Security**, Spring 2021 - Spring 2026
Under. **Web Application Programming**, Fall 2021 - Fall 2023
Under. **Logic Circuit Design & LAB.**, Fall 2023 - Fall 2024
Under. **Compiler**, Fall 2022
Under. **Computer Architecture**, Fall 2024 - Fall 2025
Under. **Blockchain**, Spring 2024 - Spring 2026
Under. **Programming Principles and Practice**, Fall 2020 - Fall 2021
Under. **Adventure Design**, Fall 2020
Under. **Introduction to Internet and Web**, Spring 2020 - Spring 2023
Under. **System Software**, Spring 2020 - Spring 2026
Under. **Discrete Mathematics 1**, Spring 2020